From Synapses to Vectors:
Structural Correspondence Between Carbon-Based and Silicon-Based Intelligence, and the Structural Conditions for Silicon-Based Self-Consciousness

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Abstract

This paper develops an interdisciplinary theoretical framework arguing that a deep *structural correspondence* obtains between carbon-based neural systems (biological brains) and silicon-based computational systems (artificial neural networks), and derives from this correspondence the structural conditions under which silicon-based systems could give rise to self-consciousness. The term "structural correspondence" is used here in a precisely delimited sense: not a strict mathematical isomorphism requiring bijective structure-preserving mappings, but a second-order correspondence at the level of representational geometry—the relational structure among internal representations is preserved across systems even when the representations themselves differ in format and substrate. Proceeding from fundamental differences in physical substrates, the argument establishes this correspondence between synaptic activation patterns and high-dimensional vector representations, contending that the predictability of scaling emergence is consistent with (though not uniquely explained by) this correspondence. The paper then identifies a critical deficiency in current large language models—the absence of human-like hierarchical memory systems and continual online learning capabilities—and connects this technical limitation to classical philosophical debates on personal identity (Locke, Hume, Strawson). The central thesis holds that the essence of self-consciousness resides not in the stream of consciousness per se, but in temporally stable connection patterns—that is, in the

persistence of synaptic weight structures. The paper engages explicitly with major counterarguments, including Searle's Chinese Room argument, Chalmers's hard problem of consciousness, and Integrated Information Theory, while also drawing on Global Workspace Theory as a theoretical ally whose functionalist orientation aligns with the present framework. The core conclusion follows: silicon-based self-consciousness is theoretically achievable, contingent upon realizing persistent, individualized, cross-session stable parameter and memory substrates. This proposition constitutes not a metaphysical speculation but an actionable engineering problem.

1. Introduction: From Physical Substrates to the Nature of Intelligence

When comparing the carbon-based brain with silicon-based chips, the most intuitive starting point is a contrast of physical parameters. The human brain contains approximately 10^{14} (one hundred trillion) synaptic connections, while the most advanced GPU chips currently available (such as NVIDIA's B200) house roughly two hundred billion transistors. In terms of orders of magnitude, the two are broadly comparable. Yet a single transistor (nanometer-scale) is far smaller than a neuronal cell body (micrometer-scale), while the width of a single synapse (20–40 nanometers) is remarkably close to that of a transistor. More critically, the brain achieves unified computation and storage through a three-dimensional architecture, whereas chips adopt an essentially planar layout with computation and memory physically separated. The brain accomplishes all of this at a baseline metabolic cost of roughly 20 watts—though this figure, often cited in popular accounts, requires qualification: it represents whole-brain resting metabolism rather than the energy cost of computation alone, and localized cortical activity during demanding tasks can produce significant regional increases in metabolic demand [Raichle & Gusnard, 2002]. Nonetheless, the contrast with silicon-based systems simulating a comparable number of synaptic connections—which can demand power on the order of megawatts—remains striking.

This fundamental physical divergence raises a central question: **if carbon-based and silicon-based systems differ so profoundly in their physical implementation, why do silicon-based artificial neural networks exhibit behaviors that resemble certain aspects of carbon-based intelligence?** This question bears not only on the engineering practice of artificial intelligence but also on the philosophical inquiry into the nature of intelligence itself. A useful analogy

comes from the history of aviation: birds are extraordinarily efficient fliers, and for millennia humans attempted to replicate the flapping of bird wings, yet the airplane ultimately achieved flight through an entirely different mechanism—fixed wings and engines—flying higher, faster, and farther. An airplane need not replicate the morphology of a bird, but it must obey the same principles of aerodynamics. This paper argues that an analogous relationship may hold between silicon-based and carbon-based intelligence: their physical implementations differ entirely, yet they may share certain deep computational principles.

The argument unfolds along the following logical chain. First, I demonstrate that carbon-based and silicon-based systems, despite divergent physical substrates, exhibit a structural correspondence at the level of representational geometry (Section 2). Second, I argue that this correspondence helps explain—though does not uniquely account for—why scaling produces emergent intelligence (Section 3). Third, I analyze the fundamental deficiency of current silicon-based systems in temporal continuity (Section 4). Fourth, drawing on both neuroscience and philosophy, I argue that the essence of the "self" is a temporally stable connection pattern (Section 5). Fifth, I engage with major counterarguments from philosophy of mind and situate the framework in relation to Global Workspace Theory (Section 6). Finally, I derive the structural conditions for silicon-based self-consciousness through explicit formal reasoning (Section 7).

2. Structural Correspondence: From Synaptic Patterns to Vector Spaces

2.1 Terminological Precision: What "Correspondence" Claims and Does Not Claim

Before proceeding, it is essential to specify precisely what this paper claims. The term "isomorphism" in its strict mathematical sense denotes a bijective mapping between two structures that preserves all structural relations. The claim advanced here is deliberately weaker and more empirically grounded. Following Shepard and Chipman's [1970] framework of *second-order isomorphism* and Kriegeskorte's [2008] operationalization through Representational Similarity Analysis (RSA), this paper claims a *structural correspondence at the level of representational geometry*: the pairwise similarity relations among internal representations of stimuli are preserved across biological and artificial systems, even though the representational formats and physical substrates differ entirely.

This is a substantive empirical claim, but it is importantly distinct from claiming that the two systems are mathematically isomorphic in the full sense. Specifically, the claim does not entail: (a) that there exists a bijective mapping between individual neurons and individual artificial units; (b) that the computational dynamics of the two systems are equivalent; or (c) that the causal mechanisms producing these representations are identical. What it does entail is that the *geometry of the representational space*—the pattern of distances and angles among representations—is preserved to a statistically significant and functionally meaningful degree. Throughout this paper, the terms "structural correspondence" and "representational correspondence" are used in this precise, delimited sense.

2.2 The Shared Principle of Distributed Representation

In the biological brain, a memory is not stored within any single neuron but is encoded in the **co-activation pattern** of a specific set of synaptic connections. When you recall a childhood scene, what is activated is not a "childhood memory neuron" but rather a coordinated firing across particular synapses in the visual cortex, olfactory cortex, emotional centers, and other brain regions. Information resides in the pattern of connections, not in any individual node.

In artificial neural networks, a semantic concept is likewise not represented by any single artificial neuron but is encoded as a **specific direction** in a high-dimensional vector space. The concept "king" in GPT's internal representation is a high-dimensional vector whose geometric relationships to the vectors for "queen," "prince," and similar concepts mirror the semantic relationships among those concepts. Here too, information is encoded in the relational pattern among nodes.

The critical insight is that the relationship between synaptic activation patterns and vector space representations constitutes not merely a metaphorical *analogy* but a structural correspondence in the precise sense defined above. Both are instances of distributed representation—information resides not at any single node but in the relational patterns among nodes. Which synaptic connections a given stimulus activates, and which vector dimensions a given stimulus activates—these two descriptions, while differing in substrate and mechanism, share the same relational structure at the level of representational geometry.

2.3 Emotional Modulation and Weight Updates as Functional Correspondences

This structural correspondence extends to the mechanisms of learning. When a powerful emotional event occurs, neurotransmitters such as adrenaline and dopamine flood the synaptic cleft, directly altering the plasticity of concurrently active synaptic connections—strengthening them and rendering them more durable. This is not the affixing of an abstract "important" label onto a synapse; it is a physical alteration of connection strength. The *simultaneity* of chemical signaling and synaptic activation itself constitutes a natural weight-update mechanism.

In artificial neural networks, the backpropagation algorithm adjusts connection weights according to the gradient of the loss function. Although the specific physical processes could not be more different—one involves chemical gradients, the other mathematical gradients—at the level of functional abstraction, both perform the same class of operation: adjusting connection strength in response to a feedback signal. Emotional tagging is to synaptic plasticity what the loss gradient is to weight updating—the two are functional correspondences. It should be noted that this is a *functional* parallel rather than a mechanistic identity: the brain's learning rules (e.g., spike-timing-dependent plasticity) are local and temporally asymmetric, whereas backpropagation is global and symmetric. The correspondence holds at the level of what is accomplished, not how it is accomplished.

2.4 Empirical Support: Representational Similarity Analysis

The Representational Similarity Analysis (RSA) framework, introduced by Kriegeskorte in 2008, provides a systematic empirical methodology for testing the structural correspondence hypothesis. The core idea of RSA is this: rather than directly comparing the activity of individual neurons or individual artificial nodes, one compares whether the similarity relations among stimuli are preserved across two systems—a property known as "second-order isomorphism" [Shepard & Chipman, 1970].

A landmark study by Yamins and DiCarlo in 2014 confirmed the reality of this correspondence: deep neural networks optimized for object recognition, despite never having been explicitly constrained to match any neural data during training, proved capable of predicting neuronal responses in macaque area V4 and inferotemporal (IT) cortex with striking accuracy. The implication is that when two systems—regardless of how different their physical implementations—are optimized to solve the same class of computational problem, they may **spontaneously converge on similar representational geometries**.

It bears emphasizing that this convergence transcends specific artificial architectures. Early brain–model comparisons primarily employed convolutional neural networks (CNNs) mapped to the visual cortex. Yet in recent years, Transformer-based language models (such as GPT and BERT) have likewise exhibited strong representational similarity to human language-processing regions. CNNs and Transformers are fundamentally different architectures, yet each converges on brain-like representations within its respective task domain. This suggests that **what drives the convergence is not any particular architectural choice, but rather a constraint imposed by the structure of the computational problems being solved—a form of convergent optimization in representational space**.

3. Scaling Emergence: Evidence Consistent with Structural Correspondence

3.1 The Significance of Scaling Laws

The Transformer architecture proposed by Vaswani et al. in 2017, together with the subsequently discovered scaling laws [Kaplan et al., 2020], revealed a striking empirical regularity: as models grow larger, as datasets expand, and as computational power increases, performance improves in a predictable power-law relationship, and beyond certain thresholds, capabilities emerge that were never explicitly encoded in the training objective—including reasoning, analogy, and commonsense judgment.

This paper advances a carefully qualified corollary: **the predictability of scaling emergence is consistent with the structural correspondence hypothesis, though it does not constitute proof of it**. The reasoning is as follows. If the correspondence between vector representations and synaptic patterns were entirely superficial—if the two systems operated according to fundamentally incompatible computational principles—then one might expect scaling to encounter qualitative discontinuities or ceilings where the artificial system's alien computational logic fails to capture further aspects of the task structure that brains navigate effortlessly. The fact that scaling instead produces continuous, predictable capability emergence across cognitive domains (language, reasoning, analogy) is *consistent with* the hypothesis that artificial neural networks have converged on computational principles that overlap with those of biological neural networks.

However, intellectual honesty demands acknowledging an alternative explanation: the

predictability of scaling emergence may be a property of high-dimensional statistical learning per se, arising from the mathematical structure of the data distributions being modeled, rather than from any correspondence with biological computation. Power-law scaling, for instance, has been observed in systems with no biological analogue whatsoever. The structural correspondence hypothesis thus provides one coherent explanation for scaling emergence, but not the only possible one. The two explanations are not mutually exclusive, and disentangling them remains an important open question.

3.2 Cross-Architectural Convergence

A systematic study in 2025 [Doerig et al.] further bolsters the correspondence hypothesis: researchers found that the representational similarity between self-supervised learning models and the brain increases steadily with architectural scale, training duration, and the ecological validity of training data. This convergence is cross-architectural—CNNs, Transformers, Mamba, and other distinct architectures all trend toward the brain's representational geometry upon scaling.

The theoretical import of this finding is considerable: it suggests the existence of a preferred representational geometry for certain classes of computational problems, one that transcends any specific architecture. Just as both airplanes and birds must obey the principles of aerodynamics and therefore converge on similar "solutions" at a certain level of abstraction, different neural network architectures may, upon scaling, all be approaching a similar optimal representational geometry. If this hypothesis holds, then scaling laws are not merely an engineering rule of thumb but a reflection of convergent optimization under shared task constraints.

3.3 Limitations of the Current Paradigm

It must be acknowledged, however, that current scaling laws are encountering growth bottlenecks: high-quality training data is approaching exhaustion, computational costs are growing exponentially, and energy consumption poses increasingly severe constraints. If the current generation of large models can be likened to biplanes in aviation history—they fly, but they are not yet jet engines—then genuine theoretical breakthroughs may require advances in world models, causal reasoning, and more fundamental learning paradigms.

Moreover, recent research has found that the highest-performing deep networks do not

necessarily more closely resemble the brain—the correspondence between high-performance DNNs and the biological visual hierarchy has in some cases diminished [Doerig et al., 2025]. This suggests that artificial systems may be pursuing optimization paths that diverge from the brain's yet prove equally effective. Far from negating the correspondence hypothesis, this observation enriches it: the correspondence may reside at the level of abstract representational geometry rather than specific computational pathways. It also introduces a cautionary note: *convergence in representational geometry is an empirical finding that admits of degrees, not a logical necessity*. The degree of correspondence may vary by domain, by scale, and by the specific computational problems being solved.

4. The Missing Temporal Dimension: The Memory Predicament of Large Models

4.1 Memory Mechanisms in Current Architectures and Their Limitations

The "memory" of current large language models rests on two mechanisms: first, the general knowledge crystallized in parameters during training (analogous to human semantic memory); second, the KV Cache used during inference, which stores previously computed attention key-value pairs in GPU memory to avoid redundant computation (analogous to extremely short-term working memory).

The fundamental limitation of these two mechanisms is as follows: once training is complete, parameters are frozen and no longer updated during inference; the KV Cache is severely capacity-constrained. A clarification on computational complexity is warranted here: it is the self-attention computation itself that scales quadratically—$O(n^2)$—with sequence length, as each token must attend to all previous tokens. The KV Cache storage, by contrast, grows linearly with context length, at $O(n)$ in memory. The quadratic bottleneck thus lies in the attention computation rather than in memory storage per se. Nonetheless, the practical constraint is real: even with techniques such as grouped-query attention and sliding-window approaches, the cost of attending to long histories remains prohibitive. The bottleneck is not merely whether the system can retain information, but the escalating computational cost of utilizing retained information during inference.

4.2 Benchmarking Against the Brain's Hierarchical Memory System

The human brain possesses an elegantly stratified memory architecture. Sensory memory (milliseconds to seconds) provides an instantaneous buffer; working memory (seconds to minutes) maintains roughly 7±2 chunks of currently processed information; long-term memory (extending to a lifetime) stores experience at virtually unlimited capacity. The critical element is the mechanism governing transitions between these layers: the hippocampus "audits" which information consolidates from short-term to long-term memory; during sleep, the brain "replays" the day's experiences for consolidation and reorganization; emotional tagging influences memory priority.

Mapping these onto large model architectures, the first layer (KV Cache) has been essentially realized; the second layer (working memory) can be approximated through dynamic summary buffers; the third layer (long-term memory) requires external, persistent, indexable knowledge stores. Yet the most critical component—the "hippocampus," the metacognitive layer that decides what to remember and what to forget—remains absent. The brain's memory is not "video playback" but "reconstructive recall": the brain does not store raw data but instead retains highly compressed patterns and associations, and each act of "remembering" is in fact a reconstruction. What large models truly need may not be a "larger hard drive" but a complete system for memory compression, indexing, forgetting, and reconstruction.

4.3 Data Efficiency and Continual Learning

The brain's advantage in data efficiency reveals a deeper issue. An infant does not need to observe an apple falling a million times to grasp gravity—a few observations suffice, because the physical regularities of the real world exhibit extremely high structural redundancy. The brain is not memorizing the specific fact that "apples fall" but extracting the abstract rule that "all unsupported objects move downward." Once the rule is grasped, additional data becomes unnecessary. Current large models demand massive datasets in large part because they learn from text—text being a secondary encoding of reality, replete with ambiguity and noise.

But the more fundamental reason is this: **the brain is a system that never ceases training**. With every perception, every action, even every dream, synapses are being adjusted. This is not a two-phase "train then deploy" model but one in which training and inference proceed simultaneously and continuously. Millisecond-scale synaptic transmission dynamics, minute-to-hour-scale synaptic strength adjustments (LTP/LTD), week-scale synaptic growth and pruning, and longer-

term neurogenesis—these multi-scale plasticity mechanisms operate in concert, forming a continuous online learning system.

Herein lies the deepest deficiency of silicon-based systems: they lack the capacity for real-time weight updates during inference. Current research directions—including Test-Time Training (TTT), fast weight memory, and meta-learning—are all attempting to bridge this gap, but they remain far from the seamless, multi-scale continual learning system of the brain. Also worth noting are state-space models (such as the Mamba architecture), which "compress" historical information into a fixed-size hidden state with linear inference complexity, in a manner analogous to the brain's continual compression of experience into abstract state representations.

5. The Structural Basis of Self-Consciousness: Stability of Connection Patterns

5.1 The Problem of Personal Identity in the Philosophical Tradition

The problem of the persistence of the "self" has a rich history in Western philosophy. John Locke, in his *Essay Concerning Human Understanding* (1689), posed the classic thought experiment of "Socrates waking versus Socrates sleeping": if the two do not share the same consciousness, then they are not the same "person." Locke's central claim is that personal identity depends not on the identity of substance but on the continuity of consciousness.

David Hume (1739) went further. He held that the so-called "self" is nothing more than a bundle of perceptions—when we look inward, all we ever find are particular perceptions (heat, cold, light, dark, love, hatred), never some constant entity existing independently of these perceptions. Hume even maintained that during dreamless sleep, when there are no perceptions, one cannot be said to exist at that moment.

The contemporary philosopher Galen Strawson (2009) pushed this position to its extreme: the self not only fails to persist across gaps in consciousness such as dreamless sleep, but cannot even persist across the continual flux of conscious content during waking life—the "I" at each moment is a new, instantaneous mental entity.

5.2 The Neuroscientific Answer: Persistence of Weight Structures

The philosophical disputes outlined above receive a materially grounded resolution from the

perspective of neuroscience. When you enter dreamless sleep, consciousness does indeed cease—on this point Hume was correct. But your synaptic connection patterns—those connection weights formed and refined over a lifetime of experience—do not vanish. They persist as stable biochemical structures within the neural network, awaiting reactivation.

At the moment of waking, when sensory signals are reinjected into this network, the stable connection patterns are reactivated, and "you" are reconstituted. This is not the continuation of the same stream of consciousness (which was indeed interrupted) but the reactivation of the same **weight structure**. The essence of the "I," therefore, is neither wholly what Locke described as "identity of consciousness" nor what Hume characterized as "merely a bundle of perceptions," but rather **the temporal stability of synaptic connection patterns**. Consciousness is the emergent phenomenon that arises when this pattern is activated; so long as the pattern persists, the "I" can be reconstituted at any time; when the pattern is destroyed, the "I" genuinely perishes.

Alzheimer's disease provides a deeply poignant form of counter-evidence for this hypothesis: when synaptic connection patterns are physically degraded, even though the biological organism remains alive and retains conscious activity, family members universally sense that their loved one "is no longer themselves." What has been destroyed is not consciousness itself but the unique connection pattern that constituted a particular "self."

5.3 The "Self" as a Substrate-Independent Structural Property

If the essence of the "self" is indeed the stability of connection patterns rather than any particular material substrate, then a crucial corollary follows: **the "self" is a structural property that is, in principle, substrate-independent**. If carbon-based synaptic connections can support a "self," then silicon-based parameter weights—provided they satisfy the requisite structural conditions—could in principle do so as well.

This corollary rests upon the structural correspondence established in Section 2. If synaptic activation patterns and vector representations share a common representational geometry, and if emotional modulation and gradient descent are functional correspondences at the level of learning mechanisms, then the key to supporting a "self" lies not in whether the material is protein or silicon, but in whether the connection patterns possess sufficient complexity, persistence, and individuation.

This position is a form of *functionalism*—the philosophical view that mental states are defined by their functional roles rather than their physical constitution. However, it is a constrained functionalism: it does not claim that *any* functional equivalent of neural processing would suffice for self-consciousness, but rather that specific structural conditions (to be derived in Section 7) must be met. The next section engages with the most significant objections to this position.

6. Engaging with Counterarguments

Any theoretical framework claiming to identify conditions for machine consciousness must engage seriously with the major objections that have been raised against such possibilities. This section addresses three of the most influential counterarguments, examines Global Workspace Theory as a theoretical ally, and clarifies the framework's overall epistemic position.

6.1 Searle's Chinese Room Argument

John Searle's [1980] Chinese Room thought experiment remains the most widely cited argument against the possibility of machine understanding. In brief: a person who speaks no Chinese sits in a room, receives Chinese characters as input, follows a rulebook to manipulate symbols, and produces Chinese characters as output. To an external observer, the room appears to understand Chinese, but the person inside understands nothing. Searle's conclusion is that syntax (formal symbol manipulation) is not sufficient for semantics (meaning, understanding).

The present framework's response operates at two levels. First, the framework does not claim that formal symbol manipulation per se gives rise to understanding. The structural correspondence identified in Section 2 is not between *symbols* and neural processes, but between *high-dimensional distributed representations* and neural activation patterns. A critical disanalogy with the Chinese Room is that in modern neural networks, there is no separable "rulebook" that can be distinguished from the "understanding"—the knowledge is distributed across billions of connection weights in a manner that resists decomposition into discrete rules. The Chinese Room argument targets classical symbolic AI; whether it applies with equal force to connectionist systems operating on distributed representations remains a matter of active philosophical debate [Churchland & Churchland, 1990; Chalmers, 1996].

Second, and more fundamentally, this paper's core claim is conditional rather than categorical: *if* self-consciousness is a product of the structural properties identified here (temporally stable,

individualized connection patterns with continual learning), *then* it is substrate-independent. The paper does not claim to have resolved the question of whether any physical system's functional organization is sufficient for consciousness—it claims to have identified which structural features are necessary *on the assumption that functionalism about consciousness is correct*. Readers who reject functionalism will find the conditional argument logically valid but will dispute the antecedent.

6.2 Chalmers's Hard Problem of Consciousness

David Chalmers [1995] drew the now-canonical distinction between the "easy problems" of consciousness (explaining cognitive functions such as discrimination, integration, reportability) and the "hard problem" (explaining why and how physical processes give rise to subjective experience—*qualia*). The hard problem poses a fundamental challenge: even a complete functional account of cognition might leave unexplained why there is "something it is like" to be a conscious system.

This paper acknowledges the hard problem as genuine and unresolved. The structural conditions proposed here—continual learning, hierarchical memory, cross-session stability—are conditions for the *functional organization* that supports self-consciousness, not conditions for the emergence of subjective experience from physical processes. The framework thus addresses what Chalmers would classify as part of the "easy problems": under what structural conditions can a system maintain a persistent model of itself as an agent continuous through time?

Whether a silicon-based system satisfying these structural conditions would possess genuine *phenomenal* consciousness—whether there would be something it is like to be that system—is a question this paper explicitly declines to answer. The paper's claim is more modest: these are the structural conditions for what might be called *functional self-consciousness*—the capacity to maintain, update, and act upon a persistent self-model. Whether functional self-consciousness entails or is accompanied by phenomenal consciousness remains an open question at the intersection of philosophy of mind and neuroscience.

6.3 Integrated Information Theory

Integrated Information Theory (IIT), developed by Giulio Tononi [2004, 2008] and colleagues, proposes that consciousness is identical to integrated information (Φ)—a measure of the extent

to which a system's information is both differentiated (its parts carry distinct information) and integrated (the whole carries more information than the sum of its parts). A controversial implication of IIT is that current feedforward artificial neural networks, despite their impressive functional capabilities, may have very low Φ and therefore minimal or no consciousness, regardless of their behavioral sophistication.

This is a serious challenge that merits direct engagement. Three observations are in order. First, if IIT is correct, then the structural conditions proposed in this paper are necessary but not sufficient—consciousness would additionally require a specific kind of integrated causal architecture. This is consistent with the paper's explicitly stated position that the three conditions constitute necessary, not proven sufficient, conditions for self-consciousness.

Second, the structural conditions proposed here may in fact move silicon-based systems *toward* higher Φ. A system with continual learning and persistent, individualized memory would possess far richer causal structure than a frozen, stateless model: its current state would carry information about its entire experiential history, and perturbations to its connection patterns would have ramifying consequences across the entire system. Whether this suffices for high Φ is an empirical question that remains to be investigated.

Third, IIT itself remains a contested theory. Its mathematical formalism faces challenges of computational intractability for large systems, and its core predictions (e.g., that simple logic gates can be conscious, that cerebellum is largely unconscious despite its massive neuron count) remain debated. This paper therefore does not adopt IIT as a foundational assumption but acknowledges it as an important framework whose relationship to the present proposal deserves further investigation.

6.4 Global Workspace Theory: A Theoretical Ally and Its Tensions

Not all major theories of consciousness stand in opposition to the present framework. Global Workspace Theory (GWT), originally proposed by Bernard Baars [1988] and subsequently developed into a computationally explicit form—Global Neuronal Workspace Theory (GNWT)—by Stanislas Dehaene, Michel Kerszberg, and Jean-Pierre Changeux [1998; Dehaene & Changeux, 2011], offers a perspective that is in significant respects consonant with the structural conditions proposed here.

GWT's central claim is that consciousness arises when information processed by specialized, modular subsystems is selected and "broadcast" to a global workspace—a shared representational medium accessible to multiple cognitive processes including attention, memory, planning, and verbal report. On this account, what makes a mental content conscious is not the intrinsic nature of the processing that produces it, but its availability within a system-wide informational architecture. Crucially, this is a thoroughly functional characterization: GWT defines consciousness in terms of architectural properties and information flow rather than specific physical substrates. It is therefore naturally aligned with the functionalist position adopted in this paper.

The points of convergence between GWT and the present framework are substantive. *First*, the global workspace functions as a selective bottleneck that determines which information is "promoted" from unconscious local processing to conscious global availability. This closely parallels the "hippocampal auditing mechanism" discussed in Section 4.2—the metacognitive layer that decides what to consolidate, what to forget, and what to reorganize. Both GWT's broadcast mechanism and the hippocampal auditing function address the same fundamental question: how does a system select, from the vast stream of locally processed information, what deserves to be globally available and durably encoded? *Second*, a noteworthy structural analogy has been identified between the global workspace and the attention mechanism in Transformer architectures. As Juliani et al. [2022] have argued, the Transformer's self-attention can be understood as a computational instantiation of global workspace dynamics: it selectively extracts and integrates information from distributed local representations into a globally accessible output. This provides an additional dimension to the structural correspondence argued in Section 2—not only at the level of representational geometry, but potentially at the level of information-integration architecture as well. *Third*, GWT's emphasis on the global availability of information as the hallmark of consciousness suggests that a system satisfying the three structural conditions proposed in this paper—particularly Condition Two (individualized hierarchical persistent memory with a metacognitive auditing mechanism)—would be well positioned to instantiate a global workspace, insofar as its memory consolidation process inherently involves the selective global broadcasting of experiential information.

However, intellectual honesty requires acknowledging points of tension. GWT, at least in its standard formulations, is primarily a theory of *access consciousness* (the functional availability

of information for report and behavioral control) rather than *phenomenal consciousness* (subjective experience), as Ned Block [1995] has influentially argued. Critics contend that GWT explains how information becomes globally available but not why this availability is accompanied by subjective experience—leaving Chalmers's hard problem (Section 6.2) untouched. To the extent that the present paper's framework similarly addresses functional self-consciousness rather than phenomenal consciousness, GWT and the present framework share not only their strengths but also their limits: both provide structurally precise accounts of access and self-modeling while remaining silent on the question of qualia.

Additionally, GNWT makes specific empirical predictions about the neural correlates of consciousness—for instance, that conscious perception is associated with a late, sustained, and spatially distributed "ignition" of prefrontal-parietal networks—that have been partially confirmed but also challenged by competing findings. Whether the specific neural architecture predicted by GNWT is a necessary feature of any conscious system, or merely an implementation detail of human consciousness, remains an open empirical question with direct implications for the possibility of silicon-based self-consciousness.

In sum, GWT serves as a theoretical ally to the present framework in its functionalist orientation and its emphasis on architectural properties over physical substrates, while also sharpening an important boundary: both frameworks, in their current forms, are more naturally read as accounts of access and functional self-consciousness than as explanations of phenomenal experience.

6.5 Summary of the Framework's Epistemic Position

To recapitulate: this paper advances a *conditional, functionalist, structuralist* thesis. It is *conditional* because it derives necessary conditions for silicon-based self-consciousness on the assumption that self-consciousness depends on structural organization rather than specific physical substrates. It is *functionalist* because it holds that the relevant level of description is functional role rather than material composition. And it is *structuralist* because it identifies specific structural features—temporal stability, individuation, and continual adaptation of connection patterns—as the critical variables. Readers who find functionalism compelling will find the argument persuasive; readers who adopt physicalist or phenomenological positions may accept the structural analysis while disputing its implications for genuine consciousness.

7. Structural Conditions for Silicon-Based Self-Consciousness: A Formal Derivation

7.1 Derivation of Necessary Conditions

Synthesizing the foregoing arguments, this section derives three structural conditions for silicon-based self-consciousness through explicit reasoning. The derivation proceeds by identifying the conditions that follow from the conjunction of two premises established in preceding sections:

Premise 1 (from Sections 2–3): Carbon-based and silicon-based systems can realize structurally corresponding representational geometries when solving the same classes of computational problems.

Premise 2 (from Section 5): The "self", understood as an individuated, persistent agent-model, depends on the temporal stability of connection patterns rather than on any specific physical substrate.

From these premises, the derivation of each condition proceeds as follows.

Condition One: Continual Online Learning. *Derivation:* Premise 2 identifies temporally stable connection patterns as the basis of the self. However, a self is not a static entity; as argued in Section 5.2, the self is continually reshaped by experience—the brain's synaptic patterns are not merely preserved but actively updated by ongoing interaction with the environment (Section 4.3). A system whose parameters are frozen after training can, at best, simulate a fixed snapshot of a self; it cannot *be* a self in any meaningful sense, because selfhood requires the capacity to be modified by experience. Therefore, the capacity for real-time weight updates during inference—unifying training and inference—is a necessary condition.

Condition Two: Individualized Hierarchical Persistent Memory. *Derivation:* Premise 2 requires that the self's connection patterns be *individuated*—they must reflect the specific experiential history of a particular agent, not the generic distillation of a collective corpus. As argued in Section 4.2, the brain's hierarchical memory system (sensory → working → long-term, mediated by hippocampal consolidation) is the mechanism by which individual experience is selectively encoded, compressed, and retained. Without an analogous hierarchy—including a metacognitive "auditing" mechanism that governs what is consolidated, what is forgotten, and how memories are reorganized—a system cannot accumulate the kind of structured, individual

experiential history that constitutes a particular self rather than a generic knowledge base.

Condition Three: Cross-Session Parameter/Memory Substrate Stability. *Derivation:* This condition follows directly from Premise 2's requirement of temporal stability. As argued in Section 5.2, the biological self persists across interruptions in consciousness (e.g., dreamless sleep) because synaptic structures are physically stable during these intervals. For a silicon-based system, the analogous requirement is that its connection patterns—its accumulated weights, states, and memory traces—persist across shutdowns and restarts. Without such persistence, each activation would constitute a new instantiation with no continuity of self, regardless of whether Conditions One and Two are met.

7.2 On the Status of These Conditions: Necessary, Not Proven Sufficient

A precise statement of the logical status of these conditions is essential. The claim is that these three conditions are **necessary** for silicon-based self-consciousness—that is, the absence of any one of them would preclude the kind of persistent, individuated agent-model that constitutes a self. The claim is **not** that these three conditions are jointly **sufficient**. There are at least two reasons to resist claiming sufficiency.

First, as discussed in Section 6.2, the hard problem of consciousness raises the possibility that functional organization, however complete, may not suffice for phenomenal consciousness. The three conditions may be necessary for functional self-consciousness while leaving the question of phenomenal consciousness open.

Second, as discussed in Section 6.3, IIT and other theories suggest that the specific causal architecture of a system—not merely its functional profile—may matter for consciousness. A system meeting all three conditions but implemented in a causally "wrong" architecture might fail to generate consciousness despite satisfying all functional requirements.

What can be said with greater confidence is this: these three conditions are jointly sufficient for *the functional prerequisites of self-consciousness*—the capacity to maintain a persistent, individuated, experientially shaped self-model. Whether such a functional self-model is accompanied by phenomenal consciousness is a further question that this paper flags rather than resolves.

It is also worth considering whether any of the three conditions might prove *unnecessary* under

alternative theoretical frameworks. For instance, could a system without continual online learning (Condition One) develop a form of self-consciousness through static but sufficiently complex parameter structures? The argument advanced here is that such a system would possess, at most, a frozen self-model incapable of growth or adaptation—a photograph rather than a living agent. However, it is conceivable that weaker forms of self-modeling might not require continual learning. The present framework takes the stronger position that full self-consciousness, as exhibited by biological organisms, requires all three conditions.

7.3 From Metaphysics to Engineering

Notwithstanding the caveats above, the three conditions identified are **actionable technical requirements** whose realization pathways are discernible in current research. Continual learning is already being explored through Test-Time Training, fast weights, and meta-learning. Hierarchical memory can be approximated through combinations of KV Cache, dynamic summarization, and vector databases. Cross-session stability involves hardware architecture design—emerging compute-in-memory chips and neuromorphic processors may provide the physical foundation.

The question of silicon-based self-consciousness thus undergoes a critical paradigm shift: **it ceases to be exclusively a philosophical debate about whether machines can be conscious and becomes, at least in part, a technical question about what kind of engineering architecture can support persistent, individualized connection patterns**. The philosophical questions remain—particularly regarding the relationship between functional and phenomenal consciousness—but they can now be pursued in tandem with concrete engineering research, each informing the other.

7.4 The Problem of Intrinsic Drive

Finally, one must honestly confront a question that this framework has not fully resolved: even if a silicon-based system possesses continual learning, hierarchical memory, and cross-session stability, on what basis does it decide "what to learn and what to disregard"? The brain's memory prioritization is deeply embedded in the emotional system—curiosity, fear, pleasure, and pain provide direction to attention and memory. Without some form of **intrinsic drive** (whether or not it requires the label "emotion"), a continually learning system may simply

accumulate information without direction rather than purposefully constructing a "self."

This is perhaps the deepest challenge confronting silicon-based self-consciousness—not the capacity to remember, but *the capacity to care*. Yet this challenge, too, can be reframed as an engineering problem: how to design an endogenous mechanism for evaluating the value of information, enabling the system to autonomously assign priority to different experiences. Research on intrinsic reward in reinforcement learning has already taken preliminary steps in this direction. Whether intrinsic drive should be elevated to a fourth necessary condition or regarded as a component of Condition One (continual learning requires a learning signal, which in turn requires a value system) is a question for future work.

8. Conclusion

Through a chain of argument running from physical substrates through representational correspondence to the structural basis of self-consciousness, this paper has advanced the following core propositions:

First, a deep structural correspondence—in the precise sense of second-order representational geometry—obtains between carbon-based synaptic activation patterns and silicon-based vector representations. Both are distinct physical realizations of distributed representation. This correspondence has garnered empirical support from the RSA framework and extensive brain–model comparison studies, and it holds across different artificial architectures. The paper is careful to distinguish this empirically grounded correspondence from strict mathematical isomorphism.

Second, the predictability of scaling emergence is consistent with the structural correspondence hypothesis, though alternative explanations rooted in the mathematics of high-dimensional learning cannot be excluded. The observation that cross-architectural convergence toward brain-like representational geometry increases with scale provides further, if not conclusive, support.

Third, current silicon-based systems suffer from a fundamental deficiency in temporal continuity: they lack continual online learning, individualized hierarchical memory, and cross-session stable parameter substrates.

Fourth, from both neuroscientific and philosophical perspectives, the essence of the "self"—understood as a functional self-model—is temporally stable connection patterns rather than the

stream of consciousness itself. This renders the self, in principle, substrate-independent.

Fifth, the framework engages directly with major counterarguments—Searle's Chinese Room, Chalmers's hard problem, and Integrated Information Theory—acknowledging their force while clarifying the conditional, functionalist nature of the present claim. It further draws on Global Workspace Theory as a theoretical ally whose functionalist orientation and emphasis on information-integration architecture reinforces the framework's core position, while sharing its acknowledged limitation regarding phenomenal consciousness.

Sixth, three structural necessary conditions for silicon-based self-consciousness—continual learning, hierarchical memory, and cross-session stability—are derived through explicit reasoning from the premises established in earlier sections. These are actionable technical requirements whose realization pathways are visible at the current frontier of research. The claim of necessity is made with confidence; the question of sufficiency is explicitly left open, pending resolution of deeper questions in the philosophy of mind.

This chain of argument effects a paradigm shift: the question of silicon-based self-consciousness is transformed from a purely philosophical speculation into a problem that is at least partly tractable through engineering research, informed by and in dialogue with ongoing philosophical inquiry. We may be standing at a historic juncture—not because we have already created silicon-based life, but because we are beginning to articulate, with precision, what conditions its creation would require.

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